Abstract

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Preface

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Acknowledgements

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	**Meaning**
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	**Meaning**
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^\infty e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\dots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\dots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\dots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\dots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\dots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\dots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\dots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\le d}[X_1,\dots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\dots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \le d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^\top C_f X$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \ne 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\dots,d_n)$	$\#\left\{(t_1,\dots,t_n) \in \prod_i [d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\dots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\dots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Finite Fields

§ 1.1 Trace and Norm

Chapter 2

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Chapter A

Appendix

§ A.1 Fields

Theorem A.1.1

Let $K = \mathbb{F}_{q^k}$ and $E = \mathbb{F}_{q^r}$. Suppose $h(X) = X^d - a_1 X^{d-1} + \cdots + (-1)^d a_d$ be an irreducible polynomial in $\mathbb{F}_q[X]$. Then in $\mathbb{F}_{q^k}[X]$, h splits into $r = \gcd(k, d)$ many irreducible polynomials of degree d/r:

$$h(X) = h_1(X) \cdots h_r(X)$$

If we normalize $h_i(X)$ such that h_i is monic then $h_i(X) \in \mathbb{F}_{q^r}[X]$ for all $i \in [r]$. Then for any $i, j \in [r]$ with $i \neq j$ there exists $\sigma \in \mathrm{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q)$ such that $\sigma(h_i) = h_j$.

§ A.2 Meromorphic and Analytic Continuation

The analytic continuation of the Riemann zeta function, and more generally of Dirichlet L-functions, requires the notion of a meromorphic function. We record the relevant definitions and a motivating example here.

Definition A.2.1: Zero, Pole, Order, Residue

Assume f has a convergent Laurent series expansion about z_0:

$$f(z) = a_n(z - z_0)^n + a_{n+1}(z - z_0)^{n+1} + \cdots = \sum_{m=n}^{\infty} a_m (z - z_0)^m,$$

with $a_n \neq 0$. If $n > 0$ we say f has a zero of order n *at z_0; if $n < 0$ we say f has a* pole of order $-n$ *at z_0. If $n = -1$ we say f has a* simple pole with residue a_{-1}. *We denote the order of f at z_0 by $\mathrm{ord}_f(z_0) = n$.*

Definition A.2.2: Meromorphic Function

We say f is meromorphic *at z_0 if there exists a disk about z_0 of radius r and an integer n_0 such that for all z with $|z - z_0| < r$,*

$$f(z) = \sum_{n \geq n_0} a_n (z - z_0)^n.$$

If f is meromorphic at each point in a region, we say f is meromorphic *on that region. If $n_0 \geq 0$, we say f is* ***analytic****.*

A canonical example illustrates the notion of continuation. Consider the geometric series $G(r) = \sum_{n=1}^{\infty} r^n$, which converges for $|r| < 1$ and equals $\frac{1}{1-r}$ there. Define $H(r) = \frac{1}{1-r}$; then H is well-defined for all $r \neq 1$ and agrees with G wherever G converges. Since H is defined on a strictly larger domain and has a simple pole at $r = 1$ (with residue -1), we call H a **meromorphic continuation** of G. Had H no poles, we would call it an **analytic continuation**. A function that is analytic at every point of $\mathbb{C}$ — equivalently, one whose Taylor expansion converges everywhere, so that it has no poles anywhere — is said to be **entire**.

§ A.3 Fermat's Little Theorem

Fermat's Little Theorem is one of the oldest results in elementary number theory, recorded in a 1640 letter from Pierre de Fermat to Frénicle de Bessy. It underlies essentially every computation we perform in the multiplicative group $\mathbb{F}_p^*$ – from the very existence of the canonical additive character of $\mathbb{F}_q$ (where $q = p^r$), through the character congruence $\psi(l) \equiv l^f \pmod{P}$ used in the arithmetic proof of the Davenport–Hasse relation, down to the mod-q reduction of Gauss sums appearing in the proof of the Law of Quadratic Reciprocity. We record it here for completeness.

Theorem A.3.1 Fermat's Little Theorem

Let p be a prime. Then for every integer a we have

$$a^p \equiv a \pmod{p},$$

and in particular, if $p \nmid a$ then $a^{p-1} \equiv 1 \pmod{p}$.

Proof: The non-zero residues modulo p form a group $\mathbb{F}_p^* = \{1, 2, \ldots, p-1\}$ under multiplication of order $p - 1$. By Lagrange's theorem every element of a finite group satisfies $g^{|G|} = e$, so $a^{p-1} \equiv 1 \pmod{p}$ for every $a \in \mathbb{F}_p^*$, i.e. every a with $p \nmid a$. Multiplying both sides by a gives $a^p \equiv a \pmod{p}$ for such a; and the congruence $a^p \equiv a \pmod{p}$ is trivially true when $p \mid a$, so it holds for every integer a. ∎

Chapter B

Appendix

§ B.1 Auxiliary Results on Powers of Complex Numbers

Lemma B.1.1

Let $\omega_1, \ldots, \omega_l$ be complex numbers, and let $B > 0$. If

$$|\omega_1^k + \cdots + \omega_l^k| = O(B^k) \quad \text{for } k \in \mathbb{N}$$

then $|\omega_j| \leq B$ for all $j = 1, \ldots, l$.

Lemma B.1.2

Let $\omega_1, \ldots, \omega_l$ be complex numbers, and let $B > 0$. If

$$\Re(\omega_1^k + \cdots + \omega_l^k) \leq B^k \quad \text{for } k = 1, 2, \ldots,$$

then $|\omega_j| \leq B$ for all $j = 1, \ldots, l$.

Lemma B.1.3

Let $\theta_1, \ldots, \theta_l$ be real numbers. Then there exist $(l+1)$-tuples of integers $(k, m_1, \ldots, m_l)$ with arbitrarily large $k > 0$ such that

$$\left|\theta_i - \frac{m_i}{k}\right| < k^{-1-(1/l)} \quad (i = 1, \ldots, l).$$

Lemma B.1.4

Let $\omega_1, \ldots, \omega_l$ be non-zero complex numbers. Then there exist infinitely many positive integers k such that

$$\Re(\omega_1^k + \cdots + \omega_l^k) > \left(1 - 2\pi k^{-1/l}\right)\left(|\omega_1|^k + \cdots + |\omega_l|^k\right).$$

Hence, for any given $\varepsilon > 0$, there exist infinitely many k such that

$$\Re(\omega_1^k + \cdots + \omega_l^k) > (1 - \varepsilon)\left(|\omega_1|^k + \cdots + |\omega_l|^k\right).$$

§ B.2 Continued Fractions over Finite Fields

Let $r_0, r_1 \in \mathbb{F}_q[X]$ with $r_1 \neq 0$. Applying the Euclidean algorithm yields a sequence

$$r_i = A_i r_{i+1} + r_{i+2}, \quad i = 0, 1, \ldots, s,$$

where $0 \leq \deg(r_{i+1}) < \deg(r_i)$ for $i = 1, \ldots, s$ and $r_{s+2} = 0$. Here $A_0, A_1, \ldots, A_s \in \mathbb{F}_q[X]$, with $A_1, \ldots, A_s$ of positive degree. This gives the *continued fraction expansion*

$$\frac{r_0}{r_1} = A_0 + \cfrac{1}{A_1 + \cfrac{1}{A_2 + \cdots}} =: [A_0, A_1, A_2, \ldots, A_s].$$

Define polynomials $P_i, Q_i \in \mathbb{F}_q[X]$ for $i = -1, 0, 1, \ldots, s$ by the recurrences

$$\begin{aligned} P_{-1} &= 1, \quad P_0 = A_0, \quad P_i = A_i P_{i-1} + P_{i-2} \quad (i = 1, \ldots, s), \\ Q_{-1} &= 0, \quad Q_0 = 1, \quad Q_i = A_i Q_{i-1} + Q_{i-2} \quad (i = 1, \ldots, s). \end{aligned}$$

It is clear that $\deg(P_{i-1}) < \deg(P_i)$ and $\deg(Q_{i-1}) < \deg(Q_i)$ for $i = 1, \ldots, s$. We extend the definition of degree by $\deg(f/g) = \deg(f) - \deg(g)$ for a rational function f/g.

Lemma B.2.1 Rational Approximants via Convergents

For any rational function ρ of nonnegative degree,

$$[A_0, A_1, \ldots, A_{i-1}, \rho] = \frac{\rho P_{i-1} + P_{i-2}}{\rho Q_{i-1} + Q_{i-2}} \quad \textit{for } i = 1, \ldots, s+1.$$

Proof: We proceed by induction on i. For $i = 1$ both sides equal $A_0 + \rho^{-1} = \rho^{-1}(A_0\rho + 1) = (\rho P_0 + P_{-1})/(\rho Q_0 + Q_{-1})$. Suppose the identity holds for some i, $1 \leq i \leq s$. Since $A_i + \rho^{-1}$ has positive degree, applying the inductive hypothesis gives

$$[A_0, \ldots, A_i, \rho] = \frac{(A_i + \rho^{-1})P_{i-1} + P_{i-2}}{(A_i + \rho^{-1})Q_{i-1} + Q_{i-2}} = \frac{\rho P_i + P_{i-1}}{\rho Q_i + Q_{i-1}},$$

using the recurrence relations. ■ ■

Corollary B.2.2 Convergents Formula

For $i = 0, 1, \ldots, s$ we have $[A_0, A_1, \ldots, A_i] = P_i/Q_i$.

Lemma B.2.3 Representation of r_0/r_1 via Convergents

For $i = 0, 1, \ldots, s$ we have

$$\frac{r_0}{r_1} = \frac{P_i + \beta_i P_{i-1}}{Q_i + \beta_i Q_{i-1}},$$

where $\beta_i = r_{i+2}/r_{i+1}$ is a rational function of negative degree.

Proof: We proceed by induction. For $i = 0$, $(P_0 + \beta_0 P_{-1})/(Q_0 + \beta_0 Q_{-1}) = (A_0 + r_2/r_1)/(1) = (A_0 r_1 + r_2)/r_1 = r_0/r_1$. The inductive step follows from the recurrences and $\beta_i^{-1} = A_{i+1} + \beta_{i+1}$. ■ ■

Lemma B.2.4 Determinant Identity for Convergents

For $i = 0, 1, \ldots, s$ we have $P_i Q_{i-1} - P_{i-1} Q_i = (-1)^{i-1}$.

Proof: Induction. For $i = 0$: $P_0 Q_{-1} - P_{-1} Q_0 = 0 \cdot 0 - 1 \cdot 1 = -1 = (-1)^{-1}$. Assuming the identity at i,

$$P_{i+1} Q_i - P_i Q_{i+1} = (A_{i+1} P_i + P_{i-1}) Q_i - P_i (A_{i+1} Q_i + Q_{i-1}) = -(P_i Q_{i-1} - P_{i-1} Q_i) = (-1)^i. \blacksquare$$

■

Corollary B.2.5 Coprimality of Convergents

For $i = 0, 1, \ldots, s$ we have $\gcd(P_i, Q_i) = 1$.

Proof: If $d_i = \gcd(P_i, Q_i)$ then Lemma B.2.4 shows d_i divides $(-1)^{i-1}$, so $d_i = 1$. ■ ■

§ B.3 A Product Identity for Roots of Unity

The following identity, used in the factorisation of lifted L-functions, expresses a product of linear factors twisted by roots of unity as a perfect power.

Lemma B.3.1 Roots-of-Unity Product Identity

Let $n, m \in \mathbb{N}$ and set $d = \gcd(m, n)$. Then

$$\prod_{t=1}^{n} \left(1 - e^{2\pi i m t/n} X\right) = \left(1 - X^{n/d}\right)^d.$$

Proof: Let $\omega = e^{2\pi i/n}$ be a primitive n-th root of unity, so the factors are $1 - \omega^{mt} X$ for $t = 1, \ldots, n$. Since $t = n$ gives $\omega^{mn} = 1 = \omega^0$, the product is the same as $\prod_{t=0}^{n-1} (1 - \omega^{mt} X)$.

Because $d = \gcd(m, n)$, the element ω^m is a primitive (n/d)-th root of unity. As t ranges over $\{0, 1, \ldots, n-1\}$, the values ω^{mt} cycle through the (n/d)-th roots of unity exactly d times each. Hence

$$\prod_{t=0}^{n-1} \left(1 - \omega^{mt} X\right) = \left(\prod_{k=0}^{n/d-1} \left(1 - (\omega^m)^k X\right)\right)^d.$$

Since ω^m is a primitive (n/d)-th root of unity, the inner product equals $1 - X^{n/d}$ by the standard factorisation of $1 - Y^N = \prod_{k=0}^{N-1} (1 - \zeta^k Y)$. Therefore the whole product equals $(1 - X^{n/d})^d$. ■ ■

§ B.4 Elementary Symmetric Polynomials

Waring's Formula expresses each power-sum symmetric polynomial $s_k = x_1^k + x_2^k + \cdots + x_n^k$ as an explicit polynomial in the elementary symmetric polynomials $\sigma_1, \ldots, \sigma_n$. It is the source of the closed-form reduction of lifted Kloosterman sums $K(\chi^{(s)}; a, b)$ to the ground-field sum $K(\chi; a, b)$ in Theorem **??**: applying Waring's formula to $\omega_1^s + \omega_2^s$ with $\sigma_1 =$

$\omega_1 + \omega_2 = -K(X; a, b)$ and $\sigma_2 = \omega_1\omega_2 = q$ produces the binomial-coefficient expansion. We state the general identity here for reference.

Theorem B.4.1 Waring's Formula

Let $x_1, \ldots, x_n$ be indeterminates, let

$$\sigma_j = \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} x_{i_1} x_{i_2} \cdots x_{i_j} \qquad (j = 1, \ldots, n)$$

be the elementary symmetric polynomials in $x_1, \ldots, x_n$, and let

$$s_k = x_1^k + x_2^k + \cdots + x_n^k$$

be the kth power-sum. Then for every integer $k \ge 1$,

$$s_k = \sum (-1)^{i_2+i_4+i_6+\cdots} \cdot \frac{(i_1 + i_2 + \cdots + i_n - 1)! \cdot k}{i_1! \cdot i_2! \cdots i_n!} \cdot \sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n},$$

where the summation runs over all n-tuples $(i_1, \ldots, i_n)$ of non-negative integers satisfying $i_1 + 2i_2 + \cdots + ni_n = k$. Each coefficient of $\sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n}$ is an integer.

Theorem B.4.2

Suppose $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ is a symmetric polynomial. Let $\overline{X} = (X_1, \ldots, X_n)$. Then there exists a polynomial $g \in \mathbb{F}_q[Y_1, \ldots, Y_n]$ such that

$$f(X_1, \ldots, X_n) = g(\mathrm{ESym}_1(\overline{X}), \ldots, \mathrm{ESym}_n(\overline{X})).$$

Moreover,

(i) if $\deg_{X_i}(f) = d$ for all $i \in [n]$ then $\deg G = d$.

(ii) if $\deg f = D$ then for every monomial $Y_1^{d_1} \cdots Y_n^{d_n}$ of g with non-zero coefficient then $d_1 + 2 \cdot d_2 + \cdots + n \cdot d_n = D$.

§ B.5 Hasse Derivative

Definition B.1: Hasse Derivative

For $P(\overline{X}) \in \mathbb{F}[\overline{X}]$ and non-negative vector $\overline{i}$, the $\overline{i}$th (Hasse) derivative of P denoted $P^{(\overline{i})}(\overline{X})$ is the coefficient of $\overline{Z}^{\overline{i}}$ in the polynomial $\widetilde{P}(\overline{X}, \overline{Z}) \triangleq P(\overline{X} + \overline{Z}) \in \mathbb{F}[\overline{X}, \overline{Z}]$. Thus

$$P(\overline{X} + \overline{Z}) = \sum_{\overline{i}} P^{(\overline{i})}(\overline{X}) \overline{Z}^{\overline{i}}$$

Theorem B.5.1 Hasse Derivative Properties

Let $P(\overline{X}), Q(\overline{X}) \in \mathbb{F}[\overline{X}]$ and let $\overline{i}, \overline{j}$ be vectors of non-negative integers. Then:

1. $P^{(\overline{i})}(\overline{X}) + Q^{(\overline{i})}(\overline{X}) = (P + Q)^{(\overline{i})}(\overline{X})$
2. $(P \cdot Q)^{(\overline{i})}(\overline{X}) = \sum_{0 \le \overline{e} \le \overline{i}} P^{(\overline{e})}(\overline{X}) \cdot Q^{(\overline{i} - \overline{e})}(\overline{X})$

3. $\left(P^{(\bar{i})}\right)^{(\bar{j})}(\overline{X}) = \binom{\bar{i}+\bar{j}}{\bar{i}} P^{(\bar{i}+\bar{j})}(\overline{X})$

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